POWER STATIONS INFORMATION AT LAXAPANA COMPLEX FOR THE PROJECT FORMULATION OF HYDROPOWER OPTIMIZATION OF SRI LANKA

	Units	*Data*				
Name of Power Station		**Laxapana**	**New Laxapana**	**Samanala**	**WPS**	**Canyon**
Type of Generating		Hydro Power	Hydro Power	Hydro Power	Hydro Power	Hydro Power
River :						
Name of Basin		Kelani	Kelani	Kelani	Kelani	Kelani
Name of River		Kehelgmu Oya	Maskeli Oya	Maskeli Oya	Kehelgmu Oya	Maskeli Oya
Reservoir						
Catchment Area	km^2	19.42	22.65	28.48	115.2	128
High Water level	m	870.2	963.2	383	1096.4	1170.4
Low Water level	m					
Effective drawdown	m	863.8	953.4	374	1076.2	1145.44
Storage Capacity	10^3m^3	394	925.11	202.3	59700	114714
Usable Capacity	10^3m^3	245.3	629.07	113.49	53583	107929
Mean annual runoff	m^3/s					
Annual Total flow	10^6m^3					
Dam						
Type		Concrete gravity	Concrete gravity	Concrete gravity	Concrete gravity	Concrete gravity
Height	m	28.65	28	29.6	47.2	41.5
Crest Length	m	73.2	24.3	32	143.3	32
Volume	m^3	27882.13	22607	50663	102485.7	101732.1
Intake						
Type of Intake		Rotating Screen	Fixed Screen	Fixed Screen	Fixed Screen	Fixed Screen
Inside diameter	m	2.29				3.4
Length	m					65
Headrace						
Type of waterway		Horse shoe shape tunnel	U-shape tunnel	U-shape tunnel	Horse shoe shape tunnel	Horse shoe shape tunnel
Inside diameter	m	2.7	2.3	5.2		3.4
Length	m	2560	5447	7742	6096	3830
Headrace Surge tank						
Type of Surge tank		Simple type	Restricted Orifice type	Simple type	Simple type	Restricted Orifice type
Inside diameter	m	12.2	6	5	18.3	9.5
Height	m	21.77	90	36.5	36.5	70